

Advanced Econometrics

Distributed Lag Models

Ewa Weychert, PhD

Testing for Stationarity: ADF via testdf2

Dickey–Fuller hypotheses

H_0 : variable is **non-stationary** (unit root present).

H_1 : variable is **stationary**.

How to pick the correct line in testdf2 output

- 1 Find the line where **all** BG test p -values are $> 5\%$ (no residual autocorrelation of any order).
- 2 Read the ADF test statistic and p -value **from that line only**.
- 3 If p -value $< 5\% \Rightarrow$ **reject** $H_0 \Rightarrow$ variable is **stationary**.
- 4 If p -value $\geq 5\% \Rightarrow$ **fail to reject** $H_0 \Rightarrow$ variable is **non-stationary**.

```
testdf2(variable = diff(USA[,c("dpi")]), test.type = "nc", max.augmentations  
= 7, max.order = 5)
```

ADF Results: dpi and consumption

Δ dpi — conclusion

Appropriate line: **6th** (5 augmentations, all BG p -values $> 5\%$).

ADF statistic: -2.01 , p -value $= 4.4\% < 5\%$

$\Rightarrow$ **reject** $H_0 \Rightarrow \Delta$ dpi $\sim I(0)$, i.e. **dpi** $\sim I(1)$ — use first differences. ✓

Δ consumption — problem

Appropriate line: **3rd** (2 augmentations).

ADF statistic: -1.85 , p -value $= 6.5\% > 5\%$

$\Rightarrow$ **cannot reject** $H_0 \Rightarrow \Delta$ consumption appears **non-stationary**!

- Plot shows **increasing variance** in the first 10 years (1950–1959).
- Two remedies tried: (1) log-transform, (2) restrict sample to 1960+.
- After restricting to 1960+: ADF statistic -2.1 , p -value $= 3.5\% < 5\% \Rightarrow$ **stationary**. ✓

Why Restrict the Sample to 1960+?

The problem: 1950–1959

The first 10 years have **unusually large variance** — post-war recovery, structural breaks. This makes the ADF test conclude non-stationarity even when the rest of the series is stationary.

- **Diagnosis:** plot the series — large amplitude in 1950s, stable from 1960 onward.
- **Solution:** drop the first 40 observations (`USA[41:204,]`) and re-run `testdf2`.
- After restriction: $\Delta \text{consumption}$ is stationary ($p = 3.5\%$). ✓
- **Rule:** if one variable is restricted, **all** variables in the model must use the *same* sub-period.
- In `dynlm()`: `set start = c(1960, 1)`.

Alternative: log-transform

Taking log stabilises the variance. $\Delta \log(\text{consumption})$ is stationary after restricting to 1960+ — same conclusion.

Alternative: Log-Transformed DL Model

Why log-transform?

First differences of **log-transformed** variables approximate **growth rates**:

$$\Delta \ln(\text{consumption}_t) \approx \frac{\text{consumption}_t - \text{consumption}_{t-1}}{\text{consumption}_{t-1}}$$

Also stabilises **increasing variance**.

Model in R

```
USA$ln_consumption = log(USA$consumption) USA$ln_dpi = log(USA$dpi)
dl_ln1 <- dynlm(d(ln_consumption) ~ d(ln_dpi) + L(d(ln_dpi))), data =
USA, start = c(1960, 1))
```

- **Interpretation changes:** coefficients are now in **growth rates**.
- $\hat{\beta}_1$: if income grows by 1%, consumption grows by $\hat{\beta}_1\%$ *in the same period*.
- Short-term and long-term multipliers calculated **identically** to the non-transformed model.
- Use when the series shows **increasing amplitude** over time.

Why Distributed Lag Models?

Core idea

In many economic relationships the effect of a change in X on Y is **spread over time**, not instantaneous.

- A household receives extra income today — but adjusts consumption *gradually*.
- A policy change affects output — with *implementation lags*.
- Standard OLS: captures only the **contemporaneous** effect β_1 .
- DL models: capture **both** current *and* lagged effects.

Our running example

Does a change in disposable personal income (Δdpi) affect consumption ($\Delta\text{consumption}$) **immediately** or also **with a delay**?

Distributed Lag Model — General Form

General DL(p) specification

$$Y_t = \alpha + \beta_0 X_t + \beta_1 X_{t-1} + \cdots + \beta_p X_{t-p} + \varepsilon_t = \alpha + \sum_{i=0}^p \beta_i X_{t-i} + \varepsilon_t$$

- β_0 : **impact multiplier** — effect of X on Y in the *same* period.
- β_i ($i > 0$): **delayed multipliers** — effect of X_{t-i} on Y_t .
- p : lag order (chosen by AIC/BIC or theory).
- **Long-term multiplier**: $\text{LTM} = \sum_{i=0}^p \beta_i$

Our case: DL(1)

$p = 1$, one explanatory variable (Δdpi), estimated on US data 1960–2016.

The Lag Operator

Definition

The lag operator L shifts a variable one period back:

$$L(x_t) \equiv x_{t-1}, \quad L(x, p) \equiv x_{t-p}$$

First-difference notation

$d(\cdot)$ denotes the first difference operator:

$$d(\text{consumption}) \equiv \Delta \text{consumption}_t$$

$$d(\text{dpi}) \equiv \Delta \text{dpi}_t$$

DL Model Specification

Estimated equation

$$\Delta \text{consumption}_t = \beta_0 + \beta_1 \Delta \text{dpi}_t + \beta_2 \Delta \text{dpi}_{t-1} + \varepsilon_t \quad (1)$$

- β_1 : contemporaneous effect of income on consumption
- β_2 : one-period lagged effect ($L(\Delta \text{dpi})$)
- Estimated in R via `dynlm()`:

```
dl_1 <- dynlm(d(consumption) ~ d(dpi) + L(d(dpi)), data = USA, start  
= c(1960, 1)) summary(dl_1)
```

How the Data Are Structured

t	cons_t	dpi_t	Δcons_t	Δdpi_t	Δdpi_{t-1}
1960	1597	1723	—	—	—
1961	1630	1769	33	46	—
1962	1711	1844	81	75	46
1963	1780	1916	69	72	75
1964	1888	2055	108	139	72
1965	2007	2196	119	141	139

Note: Δ columns are first differences of the level columns; lagged column = Δdpi_t shifted down by one row.

Estimation Results

Empirical model

$$\widehat{\Delta\text{consumption}}_t = 12.6 + 0.379 \Delta\text{dpi}_t + 0.181 \Delta\text{dpi}_{t-1}$$

- Residual SE: 21.06 on 161 df
- $R^2 = 0.3074$, Adjusted $R^2 = 0.2988$
- F-statistic: 35.73 on 2 and 161 df, $p\text{-value} \approx 0$

F-test conclusion

H_0 : all variables (besides the constant) are insignificant.

$p\text{-value} \approx 0 < \alpha = 5\% \Rightarrow$ we reject H_0 : our model makes sense in the context of joint significance of all variables.

Interpretation of Parameters

Empirical model (reminder)

$$\widehat{\Delta\text{consumption}}_t = 12.6 + 0.379 \Delta\text{dpi}_t + 0.181 \Delta\text{dpi}_{t-1}$$

- $\hat{\beta}_1 = 0.379$ — the **short-term multiplier**:
if Δdpi increases by 1 unit, we expect $\Delta\text{consumption}$ to increase by 0.379 units *in the same time period*.
- $\hat{\beta}_2 = 0.181$ — the **one-period lagged effect**:
an additional 0.181 unit increase in consumption is expected *in the following period*. **Meaningless**
- **Long-run multiplier**: $\hat{\beta}_1 + \hat{\beta}_2 = 0.379 + 0.181 = 0.560$

Long-Term Multiplier

Short-term multiplier (reminder)

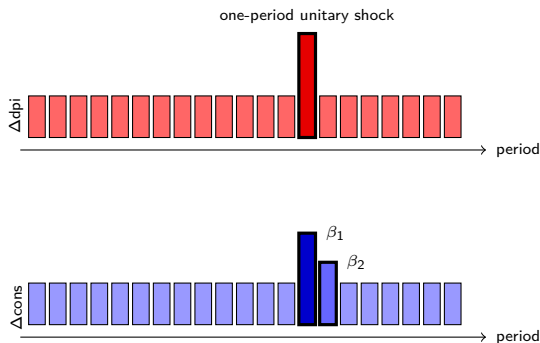
$\hat{\beta}_1 = 0.379$ — if $\Delta \text{dpi} \uparrow 1$ unit, we expect $\Delta \text{consumption} \uparrow 0.379$ units *in the same time period*.

Long-term multiplier (LTM)

$$\text{LTM} = \sum_{i=1}^{+\infty} \beta_i = \beta_1 + \beta_2 = 0.379 + 0.181 = 0.56$$

- ① If there is a **one-period unitary shock** to Δdpi , then we expect the deviations from the steady state to **sum up to 0.56**.

Impulse Response to a One-Period Unitary Shock



LTM formula

$$LTM = \beta_1 + \beta_2 = 0.56$$

- **Top panel:** Δdpi receives a shock of 1 at period 14, then returns to baseline.
- **Bottom panel:** consumption responds by β_1 at period 14 and by β_2 at period 15.
- Total deviation from steady state sums to $LTM = 0.56$.

Interpreting the Long-Term Multiplier

Short-term multiplier

$\hat{\beta}_1 = 0.379$: if $\Delta \text{dpi} \uparrow 1$ unit $\Rightarrow \Delta \text{consumption} \uparrow 0.379$ units *in the same time period*.

Long-term multiplier

$$\text{LTM} = \sum_{i=1}^{+\infty} \beta_i = \beta_1 + \beta_2 = 0.56$$

- 1 If there is a **one-period unitary shock** to Δdpi , then we expect the deviations from the steady state to **sum up to** $\text{LTM} = 0.56$.
- 2 If there is a **permanent unitary shock** to Δdpi , then we expect the difference between the new and the old steady state to be **equal to** $\text{LTM} = 0.56$.

Model Diagnostics — Overview

#	Property	Test	H_0	We want	R function
1	Autocorrelation	Breusch–Godfrey	no autocorrelation	fail to reject	bgtest()
2	Heteroskedasticity	Breusch–Pagan	homoskedasticity	fail to reject	bptest()
3	Linearity	Ramsey RESET	model is linear	fail to reject	resettest
4	Normality	Jarque–Bera	errors are normal	fail to reject	jarque.be
5	Stability	Chow	no structural break	fail to reject	sctest()

Packages: lmtest, strucchange, tseries. For a **good model** we want to **fail to reject** all five H_0 .

Model Diagnostics — Hypotheses & Intuition

1 **Autocorrelation** (Breusch–Godfrey)

H_0 : no serial correlation in residuals.

If violated: OLS standard errors are wrong $\Rightarrow$ invalid inference.

2 **Heteroskedasticity** (Breusch–Pagan)

H_0 : $\text{Var}(\varepsilon_t) = \sigma^2$ (constant).

If violated: OLS is inefficient; use robust SE.

3 **Linearity** (Ramsey RESET)

H_0 : no omitted nonlinear terms in the model.

If violated: model is misspecified.

4 **Normality of errors** (Jarque–Bera)

H_0 : residuals are normally distributed.

If violated: t/F tests unreliable in small samples.

5 **Stability** (Chow)

H_0 : coefficients are stable over time (no structural break).

If violated: a single model does not fit the full sample.

Autocorrelation: Breusch–Godfrey Test

Hypotheses

H_0 : no serial correlation up to order q (residuals are white noise).

H_1 : serial correlation of at least one order $1 \leq j \leq q$ exists.

Auxiliary regression

Regress residuals $\hat{\varepsilon}_t$ on original regressors and lagged residuals:

$$\hat{\varepsilon}_t = \alpha_0 + \alpha_1 X_t + \rho_1 \hat{\varepsilon}_{t-1} + \cdots + \rho_q \hat{\varepsilon}_{t-q} + u_t$$

Test statistic: $LM = (n - q) \cdot R_{\text{aux}}^2 \sim \chi^2(q)$ under H_0 .

- **Reject H_0** if $p\text{-value} < \alpha \Rightarrow$ autocorrelation present.
- **Advantage over DW**: works with lagged dependent variables and higher-order autocorrelation.
- In R: `bgtest(model, order = q)` from package `lmtest`.

If Autocorrelation Found: Robust Variance–Covariance

Problem

If $\text{Cov}(\varepsilon_t, \varepsilon_s) \neq 0$, the OLS variance–covariance matrix $\widehat{\text{Var}}(\hat{\beta}) = s^2(X'X)^{-1}$ is **biased** $\Rightarrow$ standard errors, t-tests and F-tests are invalid.

Remedy: HAC robust standard errors

Use the **Newey–West** (HAC) estimator:

$$\widehat{\text{Var}}_{\text{HAC}}(\hat{\beta}) = (X'X)^{-1} \hat{\Omega}_{\text{HAC}} (X'X)^{-1}$$

where $\hat{\Omega}_{\text{HAC}}$ accounts for serial correlation *and* heteroskedasticity.

- OLS **coefficients unchanged**; only SE corrected.
- In R: `coeftest(model, vcov = NeweyWest(model))` from packages `lmtest` + `sandwich`.
- Suitable when autocorrelation is *nuisance* but model structure is correct.

Remedy: Adding More Lags of Δdpi

Key idea

We add more **past values of the independent variable** Δdpi — *not* past values of the dependent variable $\Delta\text{consumption}$.

DL(1) $\rightarrow$ DL(5): step by step

$$\text{DL}(1): \Delta\text{cons}_t = \alpha + \beta_0 \Delta\text{dpi}_t + \beta_1 \Delta\text{dpi}_{t-1} + \varepsilon_t$$

$$\begin{aligned} \text{DL}(5): \Delta\text{cons}_t = \alpha + \beta_0 \Delta\text{dpi}_t + \beta_1 \Delta\text{dpi}_{t-1} + \beta_2 \Delta\text{dpi}_{t-2} \\ + \beta_3 \Delta\text{dpi}_{t-3} + \beta_4 \Delta\text{dpi}_{t-4} + \beta_5 \Delta\text{dpi}_{t-5} + \varepsilon_t \end{aligned}$$

- Each β_i answers: “*how much does a change in income i years ago affect consumption today?*”
- In R: `L(d(dpi), c(1:5))` adds lags 1 through 5 in one go inside `dynlm()`.
- **Cost:** 5 lags = 5 lost observations + 5 extra df.

DL(5): Results and Model Selection

Estimated DL(5) model in R

```
dl_5_all <- dynlm(d(consumption) ~ d(dpi) + L(d(dpi),c(1:5)),
data=USA, start=c(1960,1)) dl_5 <- dynlm(d(consumption) ~ d(dpi) +
L(d(dpi),c(2,5)), data=USA, start=c(1960,1))
```

- **DL(5) all lags:** $R^2 = 0.3451$, Adj. $R^2 = 0.3201$, F-stat 13.79 on 6 and 157 df.
- Only lags 1 (**) and current (***) are significant; lags 2–5 are not.
- **ANOVA test** (`anova(dl_5_all, dl_5)`): $H_0: \beta_3 = \beta_4 = 0$ (lags 3 & 4 jointly zero).
Both models statistically equal $\Rightarrow$ prefer **dl_5** (fewer regressors, more df).
- **BG test on dl_5:** autocorrelation *not fully resolved* in DL models — use **Newey–West HAC** standard errors instead.

ARDL Model: Adding Lags of Y and X

From DL to ARDL

DL: lags of X only.

ARDL (AutoRegressive Distributed Lag): lags of **both** Y and X .

ARDL(5,5) specification

$$\begin{aligned}\Delta \text{cons}_t = & \beta_0 + d_1 \Delta \text{cons}_{t-1} + d_2 \Delta \text{cons}_{t-2} + \cdots + d_5 \Delta \text{cons}_{t-5} \\ & + \beta_1 \Delta \text{dpi}_t + \beta_2 \Delta \text{dpi}_{t-1} + \beta_3 \Delta \text{dpi}_{t-2} \\ & + \beta_4 \Delta \text{dpi}_{t-5} + \varepsilon_t\end{aligned}$$

In R

```
ardl_5_all <- dynlm(d(consumption)
~ L(d(consumption), c(1:5)) +
d(dpi) + L(d(dpi),c(1,2,5)),
data=USA, start=c(1960,1))
```

- d_i : effect of *past consumption* on current consumption.
- β_i : effect of *past income* on current consumption.
- **Goal**: absorb remaining autocorrelation that DL alone could not remove.

ARDL(5,5): Estimation Results

Selected significant coefficients

Variable	Lag	Estimate	Std. Error	Signif.
Δcons_{t-2}	2	0.241	0.082	**
Δcons_{t-3}	3	0.252	0.074	***
Δdpi_t	0	0.301	0.052	***

- Residual SE: 19.41 on 154 df
- $R^2 = 0.4371$, Adjusted $R^2 = 0.4042$
- F-statistic: 13.29 on 9 and 154 df, $p\text{-value} \approx 0$
- Lags 1, 4, 5 of Δcons and lags 1, 2, 5 of Δdpi are **not significant** — candidates for removal.

ARDL Model Selection: ANOVA Test

Two competing models

Model 1 (ARDL 5,5 — full):

$$\Delta \text{cons}_t = \beta_0 + d_1 \Delta \text{cons}_{t-1} + \dots + d_5 \Delta \text{cons}_{t-5} + \beta_1 \Delta \text{dpi}_t + \beta_2 \Delta \text{dpi}_{t-1} + \beta_3 \Delta \text{dpi}_{t-2} + \beta_4 \Delta \text{dpi}_{t-3} + \beta_5 \Delta \text{dpi}_{t-4} + \varepsilon_t$$

Model 2 (restricted): drops insignificant lags $d_1, d_4, d_5, \beta_2, \beta_3, \beta_4$.

ANOVA result

Model	Res.Df	RSS	F	Pr(>F)
Model 1	154	58046		
Model 2	160	60647	1.15	0.3362

Conclusion

$$H_0: d_1 = d_4 = d_5 = \beta_2 = \beta_3 = \beta_4 = 0$$

$F = 1.15$, $p\text{-value} = 33.6\% > \alpha = 5\% \Rightarrow$ **cannot reject H_0** $\Rightarrow$ **Model 2 is better** (fewer regressors, more df).

The ACF Plot: What Is It?

Definition

The **Autocorrelation Function (ACF)** plot shows the correlation between a series and its own past values.

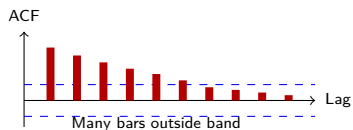
$$\rho_k = \text{Corr}(\varepsilon_t, \varepsilon_{t-k}), \quad k = 0, 1, 2, \dots$$

Each bar = correlation at one lag k .

- **X-axis:** lag k (how many periods back).
- **Y-axis:** correlation value between -1 and $+1$.
- **Blue dashed lines:** 95% confidence band.
Bars **inside** the band $\Rightarrow$ correlation is **not significant**.
Bars **outside** the band $\Rightarrow$ significant autocorrelation at that lag.
- Lag 0 is *always* 1 (a series is perfectly correlated with itself) — ignore it.

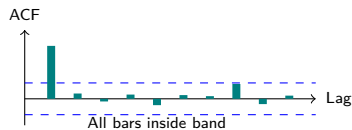
How to Read an ACF Plot

Bad: autocorrelation present



- ⇒ residuals are correlated
- ⇒ model has a problem!

Good: white noise residuals



- ⇒ residuals are independent
- ⇒ model is fine!

ACF of ARDL Residuals: Our Model

What we see in R

```
acf(ardl_5$residuals, type = 'correlation')
```

All bars at lags 1–5 lie **within** the blue dashed confidence band. Only lag 4 touches the boundary — not outside.

Confirmed by BG test

```
bgtest(residuals(ardl_5_all)~1, order = 1, 2, 3, 4, 5)
```

All p -values $> 5\% \Rightarrow$ **no autocorrelation detected.** ✓

- Adding lags of **both** Y and X (ARDL) **successfully removed** autocorrelation that DL alone could not.
- We can now trust standard errors and inference **without** needing Newey–West.
- This is why ARDL is preferred over DL when residual autocorrelation persists.

Impulse Response: Simulating a Unitary Shock

The DL model used

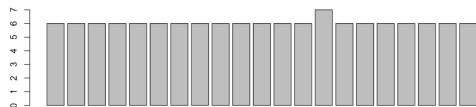
$$z_t = 1 + x_t + \frac{2}{3} x_{t-1} + \frac{1}{6} x_{t-2}$$

$$\text{LTM} = 1 + \frac{2}{3} + \frac{1}{6} = \frac{11}{6} \approx 1.83$$

R code

```
x = rep(6, times=21)  steady state = 6
z = rep(12, times=21) steady state = 12
x[14] = x[14] + 1    one shock at t=14
for(t in 14:21) {
  z[t] = 1 + x[t] + 2/3*x[t-1] + 1/6*x[t-2]
}
barplot(x); barplot(z)
```

Reading the Impulse Response Chart



- **Top panel** ($x = \text{income}$):
Flat at 6, spikes to 7 at period 14, then returns. This is the **one-period unitary shock**.
- **Bottom panel** ($z = \text{consumption}$):
Flat at 12 before shock, then:
 - $t = 14$: $+\beta_1 = +1$
 - $t = 15$: $+\beta_2 = +\frac{2}{3}$
 - $t = 16$: $+\beta_3 = +\frac{1}{6}$
 - $t \geq 17$: back to 12
- **Total deviation** from steady state
= LTM ≈ 1.83 ✓

Autocorrelation: DL vs ARDL — Why It Matters More

DL model

$$Y_t = \alpha + \beta_0 X_t + \beta_1 X_{t-1} + \varepsilon_t$$

Right-hand side contains only **exogenous** X variables.

If ε_t is autocorrelated:

only VCV is biased

$\Rightarrow$ SEs wrong, but

$\hat{\beta}$ still **consistent**.

Fix: Newey–West HAC.

ARDL model

$$Y_t = \alpha + d_1 Y_{t-1} + \beta_0 X_t + \varepsilon_t$$

Right-hand side contains **lagged** Y_{t-1} .

If ε_t is autocorrelated:

Y_{t-1} **correlates with** ε_t (endogeneity!)

$\Rightarrow \hat{\beta}$ becomes **inconsistent**.

Fix: must solve autocorrelation **structurally** (add lags).

Key rule

In **DL**: autocorrelation $\Rightarrow$ wrong SEs only — use Newey–West.

In **ARDL**: autocorrelation $\Rightarrow$ **biased and inconsistent** $\hat{\beta}$ — Newey–West is **not enough**, must fix the model.

Summary: (AR)DL Modelling Workflow

- 1 **Plot the data.** Check for trend, non-constant variance. If amplitude grows $\Rightarrow$ **log-transform**.
- 2 **Test stationarity** (`testdf2`). Pick the line where all BG p -values $> 5\%$. If non-stationary $\Rightarrow$ take first differences. If still non-stationary $\Rightarrow$ restrict sample (e.g. drop 1950–1959).
- 3 **Estimate DL model.** Start with DL(1), extend to DL(p) if needed. Drop insignificant lags using `anova()`.
- 4 **Check autocorrelation** (`bgtest`). If present in DL $\Rightarrow$ apply **Newey–West** (`vcovHAC`).
- 5 **Estimate ARDL model** if autocorrelation persists. Add lags of Y until `bgtest` fails to reject. Drop insignificant lags using `anova()`.
- 6 **Full diagnostics on final model:** autocorrelation, heteroskedasticity, linearity, normality, stability.
- 7 **Interpret:** short-term multiplier ($\hat{\beta}_1$), long-term multiplier ($\text{LTM} = \sum \hat{\beta}_i$), impulse response.

Understanding the "Cost" of Adding Lags

- **Lost Observations:** Every lag $t - k$ requires data from k periods ago. If you add 5 lags, you lose the first 5 rows of your dataset because those early dates have no "past" to reference.
- **Degrees of Freedom (df):** Each lag adds a new β coefficient to estimate. This "consumes" df , leaving less statistical "room" to find significant patterns.

The Trade-off

More lags can capture complex history but reduce the power of your tests and the size of your usable sample.

Model Selection: DL vs. ARDL

Adding lags of Y (ARDL) is often the solution when lags of X (DL) alone fail to fix autocorrelation.

Feature	DL Model	ARDL Model
Lags Included	Only Independent (X)	Both Y and X
Autocorrelation	Often persists	Successfully removed
Standard Errors	May need Newey-West	Standard OLS are reliable
Preference	Simple impact analysis	If autocorrelation persists

Table: Comparison of Distributed Lag and Autoregressive Distributed Lag Models

Final Recommendation

- **Use DL** if you only care about how past X affects Y and the residuals pass the Breusch-Godfrey test.
- **Use ARDL** if you see significant bars in your ACF plot or low p -values in BG tests after trying a DL model.
- **Check ACF**: Always inspect the Autocorrelation Function plot; "Good" residuals stay inside the blue dashed 95% confidence bands.

Thank you